

ASHUTOSH ROY

AI/ML Engineer

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SUMMARY

AI/ML Engineer specializing in **Generative AI**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, and **Speech AI**. Published researcher (**RECCAP 2026**) with research experience at **IIT Delhi**. Proficient in **Transformer Fine-Tuning**, **NLP**, **Multimodal AI**, **FastAPI**, **PyTorch**, **TensorFlow**, **LangChain**, and **MLOps**. Experienced in building and deploying **production-grade AI systems**, **intelligent agents**, and **end-to-end ML pipelines**.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built an OCR-powered document understanding pipeline for archaeological datasets; bench marked Tesseract, Marker, and Gemini OCR — selecting optimal strategy per document class to improve extraction accuracy.
- Designed hybrid retrieval (dense: FAISS/Pinecone + sparse: Apache Solr) for high-precision semantic + lexical document search over thousands of historical records.
- Deployed scalable FastAPI services for real-time LLM-based querying; integrated with frontend for sub-second retrieval responses.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi collaborators — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned transformer-based NLP models (Mistral/LLaMA variants) for healthcare applications — medical record summarization and symptom-driven conversational AI.
- Reduced patient query resolution time by 40% by deploying an AI chatbot for telemedicine with optimized intent understanding and response generation.
- Implemented GenAI-powered NLP workflows using LLMs and retrieval-augmented pipelines for intelligent healthcare automation and conversational AI systems.
- Built end-to-end ML pipelines: data preprocessing → model inference → API deployment for real-time healthcare applications.

TECHNICAL SKILLS

AI/ML & Research: LLMs, GenAI, Intelligent Automation, RAG, NLP, LangChain, Transformer Fine-Tuning (QLoRA/LoRA), Self-Supervised Learning, Multimodal AI, Speech Processing, Evaluation & Benchmarking, Dense Retrieval, Foundation Models

Libraries & Frameworks: PyTorch, Hugging Face, LangChain, TensorFlow, Scikit-learn, Pandas, NumPy, FastAPI, Streamlit, MLOps

Databases & Search: FAISS, Pinecone, Apache Solr, MySQL

Languages: Python, C++ , SQL

Tools: Git, Docker, Linux, Jupyter Notebook, Google Colab, VS Code ,OpenAI GPT, Claude, Hugging Face Transformers

PROJECTS

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Feb 2026 – May 2026

- Proposed a **self-supervised** Wav2Vec2 + BiLSTM multitask architecture for simultaneous 5-class emotion classification and continuous stress regression; achieved **85.9% accuracy**, **0.856 F1**, RMSE **0.1268**.
- Paper **accepted at RECCAP 2026, IIT Palakkad**; evaluated across 3 corpora (RAVDESS, TESS, SAVEE) with unified preprocessing (VAD, silence removal, log-Mel spectrograms).

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, LangChain, Streamlit

Dec 2024 – Apr 2025

- Built a production-ready LLM-powered contextual QA system on **LLaMA 3** with RAG; multimodal ingestion pipeline supports PDFs, images, and YouTube transcripts.
- Implemented **dense retrieval** via FAISS/Pinecone for sub-second similarity search, reducing hallucination and improving context precision.

Industrial Defect Detection System | YOLOv8, PyTorch, OpenCV, CUDA, Python

Feb 2025 – Apr 2025

- Built a real-time defect detection system using **YOLOv8** for automated quality inspection in manufacturing environments.
- Fine-tuned object detection models to identify **scratches, cracks, dents, and missing components** from production-line images.
- Implemented preprocessing, augmentation, and **GPU-accelerated inference** pipelines for robust low-latency deployment.

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR

Mar 2024 – Apr 2024

- Fine-tuned **Mistral 7B** via **QLoRA** (4-bit quantization) for domain-specific medical NLU; reduced compute cost vs. full fine-tuning while preserving task accuracy.
- Built OCR pipeline using Tesseract to extract and structure information from unstructured clinical reports for downstream LLM inference.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana